CLASS 8 — MATHS · FORMULA SHEET

Chapter 2: **Power Play** | Quick revision, one page

Laws of exponents ($a, b \neq 0$; m, n integers)

- $a^m \times a^n = a^{m+n}$ and $\dfrac{a^m}{a^n} = a^{m-n}$.
- $(a^m)^n = a^{mn}$.
- $a^m \times b^m = (ab)^m$ and $\dfrac{a^m}{b^m} = \left(\dfrac{a}{b}\right)^m$.

Zero and negative exponents

- $a^0 = 1$ for $a \neq 0$.
- $a^{-n} = \dfrac{1}{a^n}$ and $\dfrac{1}{a^{-n}} = a^n$.
- $\left(\dfrac{a}{b}\right)^{-n} = \left(\dfrac{b}{a}\right)^n$.

Standard (scientific) form

- A number is written as $m \times 10^n$ with $1 \leq m < 10$ and n an integer.
- Large numbers: positive n (e.g. $3\,50\,000 = 3.5 \times 10^5$). Small numbers: negative n (e.g. $0.0072 = 7.2 \times 10^{-3}$).
- To multiply or divide: operate on the m parts, then add or subtract the exponents of 10; re-adjust to $1 \leq m < 10$.

Useful facts

- $(-1)^n = 1$ if n is even and -1 if n is odd.
- Powers help compare very large or small quantities: compare the exponents of 10 first, then m.

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